Introduction to Data Science with R (DS002EA)

Instructor: Tyler McDaniel, Environmental Analysis

Date | Time | Location

Syllabus

Course Website: [to come]

Office Hours:

TBD, Location

Course Description:

Data science has rapidly gained recognition within the social and environmental sciences because it offers powerful new ways to ask questions. Using the R programming language, we will learn data gathering and cleaning, visualization, network analysis, modelling, visualization, ethics, and interpretation. We will explore how a range of computational methods can be used to garner new meanings from sources such as weather monitors, press releases, websites, government programs, and more. Class sessions will be taught interactively using collaborative coding tools . Students will follow along with workshop-style lectures by using and modifying code in real time to analyze data and visualize results. The course will cover such topics as the South African water crisis, Hurricane Katrina, the California Wildfires, and water rights along the Colorado River. The class will culminate with final projects in which students implement the full data science pipeline, from idea generation to results and interpretations, using a topic and data of their choosing.

Who is this course for? This course is primarily designed for people with some prior exposure to programming (such as DS1). While this course may be useful for programmers and computer scientists seeking social applications for their skills, the materials will mostly be introductory. This course is also for students who seek to develop a project related to data science and society. Over the course of sixteen weeks, students will have opportunities to pose questions and ideas, receive feedback, and collaborate with others on building out a complete research project.

Learning Goals: At the end of the course, students will have experience using data science tools to address questions related to society and the environment. We will learn what questions can be addressed by data science and consider how we can effectively convey this information with the tools that we have. We will also think critically about where data come from, how they are represented, and our role as data scientists in the twenty-first century.

Classroom policies and format: Please arrive on time and having completed the assigned materials. Being prepared to discuss the readings/videos/audio will enrich the learning experience of the entire class. Completing the assigned materials by the beginning of each week will also allow students to identify areas of interest and consider possible research questions for problem sets or for the final project.

The general class structure will be as follows:

- **Warm Up** Every day, we'll do some sort of warm-up related to the course material. This will often involve drawing and modelling data by hand.

- **Theory** We will discuss new concepts and ideas. This part of our class will be *technology free*! Why? Data science has existed long before the modern era of computing capabilities. To refine their methodological approach, researchers in the modern age often practice on small data. So we will do the same.
- **Laboratory** To implement theories of data science, students will participate in *coding exercises*. Drawing from what we learned, we will take out our laptops and devices, and try to do some (modern) data analysis ourselves using R. As with learning to code in any language, you can expect some amount of roadbumps/frustration. But we will work through this together!

Outside of class, you can expect to do the following:

- **Reading Course Materials** On your own time, your should be keeping up with course readings, notes, and other materials. These will help solidify concepts and may provide ideas for projects. I expect you to spend roughly 4 hours per week with the course materials.
- **Independent Project/Problem Set Work** On your own time, you will have a chance to implement the full data science pipeline (from ideation and data gathering to modelling and interpretation). I expect projects and assignments to take roughly 4 hours per week, although this may vary across the semester (don't put it all off until the end!).

Textbooks We will mostly use two textbooks for this course. The good news? Both are free, and online! (Though you are welcome to purchase either or both if you wish). For the rest of the Syllabus, I'll refer to these by their titles alone.

Matthew J. Salganik. 2017. Princeton University Press. Bit By Bit: Social Research in the Digital Age. This is probably the most complete book out there on doing data science for sociology (and other social sciences). The sections on data generation and research ethics are especially insightful, in my opinion.

Healy, Kieran. 2018. Princeton University Press. Data Visualization: A Practical Introduction. This is an excellent guide to working with data. It is complete with exercises and data to use in R, and it provides a welcome reminder to think seriously about the aesthetic elements of data science.

Software This class will primarily be taught in R, which is free, open access, built for and by data scientists, and has many wonderful and current packages for social scientists. You can download R here. We will use R Studio as an interface, which can be downloaded for free here. This being said, programming languages evolve and sometimes become defunct. The theoretical concepts that we cover in this class should be applicable to a range of data science programming tools. If you wish to use an alternative software, such as Python or Julia, or an alternative interface, such as Jupyter Notebook, I completely encourage this!

Assignments

- Problem Set 1 — *Due: February 1 at 11:59pm*
- Problem Set 2 — *Due: February 21 at 11:59pm*
- Problem Set 3 — *Due: March 10 at 11:59pm*
- Problem Set 4 — *Due: April 5 at 11:59pm*
- Problem Set 5 — *Due: April 29 at 11:59pm*
- Problem Set 6 — *Due: May 1 at 11:59pm*
- Problem Set 7 — *Due: May 15 at 11:59pm*
- Final Presentation — *Due: May 31 at 11:59pm*
- Final Project — *Due: June 1 at 11:59pm*

Grading

Final grades will be determined as follows:

- Participation (including in-class assignments): 25%
- Problem sets: 35%
- Final Project: 35%
- Final reflection: 5%

Participation

Coming to class prepared (having done any readings or assignments) is vital for participation. In class, students can participate in many ways. First, students can participate in classroom discussions and activities. Second, students can complete in-class coding exercises. Engaging in class activities, working well with peers and in groups, and asking relevant questions during class are all valid forms of participation. At some point in the semester, everyone will be required to share a brief visualization from their final project work.

Problem Sets

There will be seven problem sets over the course of the semester. Students are expected to complete each of these on time and turn in assignments on Canvas. Most of the assignments should be turned in as .PDF files generated by R Markdown or an alternative. Students can fork assignments and update their progress using GitHub (which serves as a useful tool for collaboration and a way to assess effort). The assignments involve synthesizing, exploring, and connecting class concepts, and should reflect student's own ideas.

Final Project, Proposal, and Presentation

The final project is intended to be an opportunity for students to apply some of the skills that we learn in this class toward a pressing and important issue. Much of this class is very introductory - we will go broad, but not deep - however, in the final project you can go deeper into one specific method or topic area.

Students can work in groups of up to *four* for the final projects. Individual projects are also fine - but I encourage you to collaborate if you have the capacity! Much of the best science is now done in teams, so this type of work is an important skill to practice. As is the norm in science, teams should write details on specifically who contributed what to the project.

The semester goes by quickly, so students should begin thinking about potential final projects early. The project proposal is due in week 4, leaving 12 weeks for students to complete the project. The project proposal should include a potential research question, data source, and method. It is likely that you will want to use one or more methods that we haven't covered yet in week 4 - this is ok! Provide as much detail as you can in your proposal.

Class Schedule and Reading List

Readings should be completed before class.

Week 1: What is Data Science? What is Computational Social Science?

We begin by exploring the meaning of "Computational Social Science," and how it relates to data science, and social science more broadly. This field has changed tremendously in the last few decades, so we'll try to orient ourselves to its history and current directions. We will load R and R Studio onto our computers and start to explore these tools.

Reading:

- Bit By Bit. 1: Introduction.
- Lazer, David; Pentland, Alex; Adamic, Lada; et al. 2009. Science. Computational Social Science.
- *Optional:* Edelmann, Achim; Wolff, Tom; Montagne, Danielle; Bail, Christopher A. 2020. Annual Review of Sociology. Computational Social Science and Sociology.
- Consider also viewing Installing R and RStudio

Week 2: "Gitting" Started: Code, Version Controls, and Reproducibility

In our second week, we'll start coding together, with purpose. We will learn about the Git system for file management, and how it is useful for version control, collaboration, and open science.

Reading:

- Excuse me, do you have a moment to talk about version control?
- Bit By Bit: Social Research in the Digital Age. Observing Behavior.
- *Optional*: How Pew Research Center uses git and GitHub for version control
- *Optional*: A layman's introduction to Git

Week 3: Gathering Data

Let's get some data! We will discuss data gathering, through processes such as text scraping and APIs, as well as collaborative data efforts. For our first foray into data gathering, we will examine trends in Google searches.

Reading:

- Bit by Bit: Social Research in the Digital Age. Asking Questions
- Bit By Bit: Social Research in the Digital Age. Creating Mass Collaboration.
- *Optional:* Hwang, Jackelyn; Sampson, Robert. 2014. American Sociological Review. Divergent Pathways of Gentrification: Racial Inequality and the Social Order of Renewal in Chicago Neighborhoods.
- *Optional:* Sugie, Naomie. 2016. Sociological Research & Methods. Utilizing Smartphones to Study Disadvantaged and Hard-to-Reach Groups.

[Week 4: Gathering (Big) Data (with SQL)]

We've now gathered some data. But what if we want to go bigger? These require a database management system, which can be queried using unique commands and language.

-

Week 5: Looking at and Restructuring Data

Now that we have some data, let's look at it! We will explore different types of data and how we might restructure them. We will also discuss types of data *invisibility* that shape which data we are able to see.

- Mobile phones give researchers a deeper look into living homeless in L.A.
- Data Visualization: A Practical Introduction. Preface.
- Data Visualization: A Practical Introduction. 1: Look at Data.
- Data Visualization: A Practical Introduction. 2: Get Started.
- *Optional:* Explore the Invisible Information Lab

Week 6: Data Visualization Techniques

Data visualization is an important, and sometimes overlooked, aspect of social science. This week, we go back to the visualization techniques of Du Bois, which remind us that data science long predates the current era of "big data."

Reading:

- Data Visualization: A Practical Introduction. 3: Make a Plot.
- Data Visualization: A Practical Introduction. 4: Show the Right Numbers.
- Data Visualization: A Practical Introduction. 5: Graph Tables, Add Labels, Make Notes.
- Battle-Baptiste, Whitney; Rusert, Britt. 2018. Princeton Architectural Press. W. E. B. du Bois's Data Portraits : Visualizing Black America.
- *Optional:* Tackett, Maria; Çetinkaya-Rundel, Mine. 2023. Chance. Analyzing and Recreating Data Visualizations of W.E.B. Du Bois.
- *Optional:* Try the Education Opportunity Explorer

Week 7: Modelling Data

Up to this point, we have mostly focused on *describing* our data. What if we want to say more about the *processes* unfolding? In this week, we look at data modelling. We will examine linear and non-linear models, their benefits, and some potential drawbacks.

- Data Visualization: A Practical Introduction. 6: Work With Models.
- *Optional:* Gelman, Andrew; Hill, Jennifer; Vehtari, Aki. 2020. Regression and Other Stories.
- *Optional:* Gelman, Andrew. 2011. American Journal of Sociology. Causality and Statistical Learning.

Week 8: Prediction and Interpretation

It's easy to understand things that have already happened, but what about future events? A new "school" of data science, which has gained popularity in recent years, seeks not just to model relationships, but to *predict* outcomes. We'll look at what this means for the social and environmental sciences.

Reading:

- Breiman, Leo. 2001. Statistical Science. The Two Cultures of Statistics.
- Molina, Mario; Garip, Feliz. 2019. Annual Review of Sociology. Machine Learning for Sociology.
- *Optional:* Surani, Faiz; Suzgun, Mirac; Raman, Vayoma; et al. 2025. AI for Scaling Legal Reform: Mapping and Redacting Racial Covenants in Santa Clara County.
- *Optional:* Johnson, Rebecca A.; Zhang, Simone. 2015. Sociological Science. Predictive Algorithms and Perceptions of Fairness: Parent Attitudes Toward Algorithmic Resource Allocation in K-12 Education.
- *Optional*: Jean, Neal; Burke, Marshall; Xie, Michaelk; et al. 2016. Science. Combining satellite imagery and machine learning to predict poverty.
- *Optional:* Bit by Bit: Social Research in the Digital Age. Running Experiments

Week 9: Maps and Spatial Data

All data are located somewhere. In this week, we consider how to work with geographical information systems (GIS) and ways to communicate the spatial dimensions of our data. We will explore some tools for interactive spatial displays.

Reading:

- El Niño's impact on aging California schools may be severe, experts warn
- Data Visualization: A Practical Introduction. 7: Draw Maps.

- Data Visualization: A Practical Introduction. 8: Refine Your Plots.
- *Optional:* Nilforoshan, Hamed; Looi, Wenli; Pierson, Emma; et al. 2023. Nature. Human mobility networks reveal increased segregation in large cities.
- *Optional:* Explore the Racial Covenant Map

Week 10: Networks

Sometimes, we wish to measure or display how observations are linked to each other. For instance, we might want to analyze a web of social interactions, or migration flows between many entities. In these times, network analysis is what we need. We will look at some applications of networks, and cover basic ways to visualize and analyze network data.

Reading:

- Network structure and influence of the climate change counter-movement
- Corbera, Esteve;, Calvet-Mir, Laura; Hughes, Hannah; Paterson, Matthew. 2016. Nature Climate Change. Patterns of authorship in the IPCC Working Group III report.
- *Optional:* Network Structure in Small Groups and Survival in Disasters

Week 11: Text Mining

Text data contains multitudes of information, and can be leveraged to do some powerful things in these times. To gear up for an expedition into text analysis, we will start by learning about text mining.

Reading:

- Canales, Kristine L.; Pope, JoEllen V.; Maestas, Cherie D. 2019. Social Science Quarterly. Tweeting Blame in a Federalist System: Attributions for Disaster Response in Social Media Following Hurricane Sandy.
- *Optional:* Baker, Dominique J., Karly Ford, Samantha Viano, and Marc Johnson-Guerrero. Racial Category Usage in Education Research: Examining the Publications from AERA Journals

Week 12: Sentiment Analysis

Reading:

- Want to Survive Climate Change? You'll Need a Good Community
- Day Zero Cape Town
- Wetts, Rachel. 2022. PNAS. In climate news, statements from large businesses and opponents of climate action receive heightened visibility

Week 13: Topic Modelling

Reading:

- As the Colorado River Shrinks, Washington Prepares to Spread the Pain
- The Colorado River is Shrinking. See What's Using All the Water

Week 14: Data Science in the AI Era

We've learned the building blocks of coding in R, but is anyone still writing code? In this week, we explore different data science "workflows," and discuss the promises and pitfalls of each.

Reading:

- Agentic AI and the next intelligence explosion
- Manning, Benjamin; Zhu, Kehang; Horton, John. 2025. Automated Social Science: Language Models as Scientist and Subjects.
- Athey, Susan; Brunborg, Herman; Du, Tianyu; Kanodia, Ayush; Vafa, Keyon. 2025. LABOR-LLM: Language-Based Occupational Representations with Large Language Models.
- Alvero, A.J.; Lee, J.; Regla-Vargas, A. et al. 2024. Journal of Big Data. Large language models, social demography, and hegemony: comparing authorship in human and synthetic text.
- Can AI Stand In for Human Survey-Takers? Not Really

Week 15: Ethics and Interpretations

Last but certainly not least, we focus on ethical considerations and data interpretation. After (nearly) a semester of learning the full data science pipeline (from ideation to interpretation), we discuss the scope of ethical issues and opportunities that face data scientists in the twenty-first century.

- Bit by Bit: Social Research in the Digital Age. Ethics
- The silicon gaze: A typology of biases and inequality in LLMs through the lens of place

Week 16: Final presentations

- No readings this week! Just work on your projects and come prepared to watch your classmates present!

A note on accessibility

I want you to come to class. If you use a name or gender that is different from what is on the registrar, let me know and I will use what you prefer. If you may need academic accommodations based on the impact of a disability, you must initiate the request with Accommodations and Academic Resource Services at 909-621-8277 or ars@scrippscollege.edu. Scripps students seeking to register for formal academic accommodations must contact ARS to formalize accommodation requests. A non-Scripps student's home campus is responsible for establishing and providing accommodations. Below you will find contact info for coordinators on the other campuses: The Student Disability Resource Center (SDRCLinks to an external site.) is the centralized resource center for support for students with disabilities at the Claremont Colleges. CM: AccessibilityServices@cmc.edu HM: access@g.hmc.edu PZ: academicsupport@pitzer.edu PO: disability@pomona.edu

A note on late assignments: I recognize that unexpected events come up, and despite our best efforts, we sometimes miss deadlines. While it is important to keep up with the class assignments, it is possible to catch up with some additional effort.

Therefore, everyone is granted *1 late assignment, no questions asked*. These can be turned in up to **two weeks** after the initial deadline with no penalty. Afterwards, or for the third late assignment, students will lose 30% of the maximum grade.

The purpose of this policy is to respect your privacy. You don't need to disclose to me *why* an assignment is late, if you don't want to. But in order for this policy to work well, it is vital that you only use these late

assignments when you really need them. I highly, highly, encourage you to submit everything on time if you are able to.

If you have any questions, or if a situation comes up where you foresee needing more than one late assignment, please come to office hours or reach out.

A note on Generative AI: The latest versions of some generative AI software have capabilities to provide human-like intuition and reasoning, and to answer novel problems in interesting ways. These tools might *supplement* our learning, but they should never *replace* our learning.

For **in class assignments**, we will learn most concepts and coding ourselves, without the help of AI. Early research suggests that it can be beneficial for higher-level tasks, but usage might limit learning of lower-level concepts. For example, one study finds that "Although entry-level developers used generative AI the most, it did not appear to benefit them." Another study notes that AI can reduce skill acquisition in math learning. For these reasons, I believe there are substantial benefits to focusing on conceptual learning, especially at the introductory level.

For **out of class assignments**, this class requires a lot of coding, problem-posing, and data interpreting. I want you to be able to do data science in the real world. We might want to ask a question like, "how do you read a csv into R?" or "how can I add spaces to my R markdown document?" from time to time. In some cases, AI can be very helpful in answering these questions, often pulling answers from sources like stackoverflow (where humans voluntarily help each other solve problems!). We will also cover coding with agents and other types of AI usage.

For each problem set, students will upload three files: (1) a markdown file, (2) a pdf, and (3) a link to their github file showing edits over time. With each of these, the teaching team will evaluate not only the finished assignment, but how the assignment was revised and changed over time. We will reward - and provide feedback on - your effort.

A note on online teaching: While I plan for a smooth and seamless semester, it is possible that there will be some disruptions to our ability to attend class together, either individually or at the collective level. To accommodate these, we may switch to a hybrid or online format for some classes.

If you are sick and do not wish to get your classmates sick, that is great. It is possible that a hybrid option will be available, but I cannot guarantee this. Therefore, you are responsible for catching up on the work that you miss.

If we move to an entirely online schedule for some classes, we will have to re-establish norms and policies. If this is the case, assume that all deadlines will remain the same.

Additional Resources

Modern Data Science with R, 3rd Edition, Baumer, Kaplan, and Horton This book is a fantastic introduction to data science, written by educators, that may be useful both as a generalist intro and as for students learning more advanced methods.

R for Data Science, 2nd Edition, Hadley Wickham & Garrett Grolemund This book is an excellent guide to using R for data science, written by the same person (Hadley Wickham) who wrote `dplyr` and many of the R functions that you will use in this class.

SICSS Learning Materials This is a wonderful trove of materials geared toward graduate students in the social sciences seeking to learn (or improve upon) their data skills. It is organized by Matthew Salganik (from Bit By Bit) and Chris Bail.